

LIKHITH NATTUVA

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SOFTWARE ENGINEER | DATA ENGINEER | CLOUD ENGINEER | PYTHON | SQL | AWS | MACHINE LEARNING

PROFESSIONAL SUMMARY

Computer Science graduate student with a strong foundation in software development, data engineering, cloud concepts, databases, Python, SQL, and machine learning. Experienced through academic and project work in building data pipelines, designing relational data models, processing structured and image datasets, developing web applications, and documenting technical workflows. Prepared for entry-level and early-career roles in software engineering, data engineering, cloud engineering, data analysis, Python development, QA automation, and application support.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C, C++, Scala, PySpark, TypeScript, JavaScript

Cloud & Data Engineering: AWS S3, AWS Redshift, AWS IAM concepts, AWS Glue concepts, ETL/ELT, data lakes, data warehouses, batch processing

Big Data & Analytics: Apache Hive, Spark DataFrames, Hadoop/HDFS concepts, partitioning, CSV, JSON, Parquet, data quality validation

Databases: PostgreSQL, MySQL, OracleDB, MongoDB, Prisma ORM, relational database design, indexing concepts, SQL Developer

Machine Learning & AI: TensorFlow, Keras, OpenCV, CNN, NumPy, Pandas, Matplotlib, image preprocessing, model evaluation

Web & Tools: React, Node.js, Express.js, REST APIs, Vite, Cucumber.js, Playwright, HTML5, CSS, Bootstrap, Django, Git, Jupyter Notebook

CS Foundations: Data structures, algorithms, object-oriented programming, operating systems, computer networks, machine learning, deep learning, NLP, computer vision

EDUCATION

Master of Science in Computer Information Science | Harrisburg University, Harrisburg, PA | Mar 2025 - Jun 2026 | GPA: 3.87

Bachelor of Technology in Computer Science Engineering | KL University, Hyderabad, Telangana, India | Jun 2019 - May 2023 | CGPA: 8.24

PROFESSIONAL EXPERIENCE

Web Design Intern | NIELIT

- Developed web pages using HTML and CSS as part of a structured internship project, translating requirements into functional page layouts.
- Built foundational experience in SDLC, debugging, web application structure, and technical documentation.
- Collaborated in a learning-focused environment to understand web development tools, page design, and project presentation.

TECHNICAL PROJECTS

AWS S3 to Redshift Data Lakehouse Project | AWS S3, AWS Glue, PySpark, Hive, Redshift, Scala, SQL, Parquet

- Designed an end-to-end batch data lakehouse workflow using S3 raw, processed, and curated zones for structured and semi-structured data.
- Applied PySpark and Scala Spark transformation logic for null handling, deduplication, type casting, joins, aggregations, and data validation.
- Created Hive external table and partitioning concepts for large file-based datasets and analytics-friendly query access.
- Modeled Redshift staging, fact, and dimension tables and prepared SQL validation checks for source-to-target reconciliation.

AWS Data Engineering Practice Project | AWS S3, Redshift, PySpark, Hive, Scala, SQL

- Designed a batch data pipeline workflow for ingesting CSV/JSON files into Amazon S3, validating schema rules, and preparing curated analytics datasets.
- Wrote Python and PySpark-style transformation logic to clean, filter, join, aggregate, and partition datasets for scalable processing.
- Documented pipeline flow, assumptions, source-to-target mapping, validation checks, and failure-handling steps for reproducibility.

Emotion-Aware Support Companion | TypeScript, React, Node.js, Express.js, PostgreSQL, Prisma, REST APIs, Gherkin, Cucumber.js, Playwright

- Developed a full-stack, text-based emotional support companion web application with conversation sessions, message analysis, supportive responses, consent management, privacy preferences, conversation history, feedback, and audit logging.
- Built a three-tier architecture using a React single-page application, Express REST API, PostgreSQL database, and Prisma ORM to manage users, sessions, messages, emotion records, crisis flags, privacy settings, and feedback entries.
- Implemented safety guardrail logic for crisis indicators and non-clinical response control, using deterministic emotion analysis and pre-authored response templates to avoid unsafe or diagnostic outputs.
- Created and validated Gherkin acceptance scenarios with Cucumber.js and Playwright, achieving 134 passing scenarios across 10 use cases, including UI, database, safety, and performance validations.

Brain Tumor Detection Using MRI Scan Images | Python, TensorFlow, Keras, OpenCV, CNN, NumPy, Pandas, Jupyter Notebook

- Built a CNN-based image classification workflow to detect whether brain MRI scan images contain a tumor or no tumor.
- Performed image preprocessing, resizing, rescaling, train/validation/test splitting, augmentation, and model evaluation using accuracy and loss metrics.
- Implemented a sequential CNN architecture with convolution, max-pooling, dropout, flatten, and dense layers, achieving approximately 94% test accuracy in the project report.

Learning Management System Web Application | Java, SQL Concepts, Web Design

- Designed an academic web application concept to manage student information, course materials, assignments, and grades.
- Applied object-oriented programming and database design concepts to organize student, course, assignment, and grade entities.

Face Mask Recognition | Python, Artificial Intelligence, Computer Vision

- Built an AI-based image/video recognition workflow to detect whether a person is wearing a face mask.
- Worked on input preparation, image/video processing, model-output evaluation, and troubleshooting for an applied computer vision use case.

CERTIFICATIONS

Fundamentals of Data Analytics | Microsoft Azure Fundamentals | Oracle Certified Foundations Associate | Aviatrix Certified Engineer - Multicloud Network Associate Course | Advanced RPA Professional | Introduction to Packet Tracer | OS Bridge Course